Scientific Justification *Be sure to include overall significance to astronomy. For standard proposals limit text to one page with figures, captions and references on no more than two additional pages.*

Quasi-Periodic Eruptions (QPEs) are intense, recurring, rapid bursts of soft X-ray radiation originating from the nuclei of galaxies, typically recurring every few hours. These events are thought to be prime candidates for the electromagnetic counterparts to Extreme Mass-Ratio Inspirals (EMRIs), stellar-mass compact objects orbiting supermassive black holes (SMBHs), which are a key source class for future low-frequency gravitational wave (GW) detectors like LISA. However, the physical origin of QPEs is fiercely debated. Competing models, such as repeating disk instabilities or accretion impacts from an orbiting body, make nearly identical predictions for their X-ray spectra and coarse timing properties.

Since the X-ray spectra are not enough, we must use the optical light curve to delineate our models. If QPEs are caused by an EMRI directly punching through the accretion disk, we will recover strictly periodic eruptions, with the quasi-periodic timing deviations (for example: from gravitational radiation or apsidal precession) occurring on predictable, long timescales. In contrast, a disk instability model, where a star orbiting a SMBH undergoes mass transfer, might showcase quasi-periodicity by default due to the nature of the mass transfer's fluid flow, with stochastic jitter or systematic period evolution on much shorter timescales. Current optical observations of QPEs are sparse and lack the continuous, high-cadence coverage needed to perform these precise timing analyses. The absence of this data is the primary obstacle to determining the nature of QPEs and, by extension, their utility as multi-messenger beacons for EMRIs.

We propose a new experiment to break this model degeneracy: obtaining the first continuous, high-cadence optical study of a QPE's light curve peaks throughout multiple eruption cycles. By measuring the arrival times of optical eruptions with minute-level precision for multiple connected cycles, we can detect or constrain period derivatives ($\dot{P}$) and timing jitter. A null detection of significant stochastic jitter or a $\dot{P}$ consistent with GW-driven inspiral would strongly favor the EMRI hypothesis. On the other hand, detection of significant jitter or an anomalous $\dot{P}$ would point toward accretion disk physics as the driver. This experiment directly addresses a fundamental question in high-energy astrophysics and while opening the door for a new EM-GW observation synergy within LISA's science case.

References

Arcodia et al. 2021, *Nature*, 592, 704 (QPE Discovery in eROSITA data)

Miniutti et al. 2023, *A&A*, 670, A93 (Detailed study of GSN 069)

Linial & Metzger 2023, *ApJ*, 957, 34 (EMRI model for QPEs)

Krolik & Linial 2022, *ApJ*, 941, 24 (Disk instability model for QPEs)

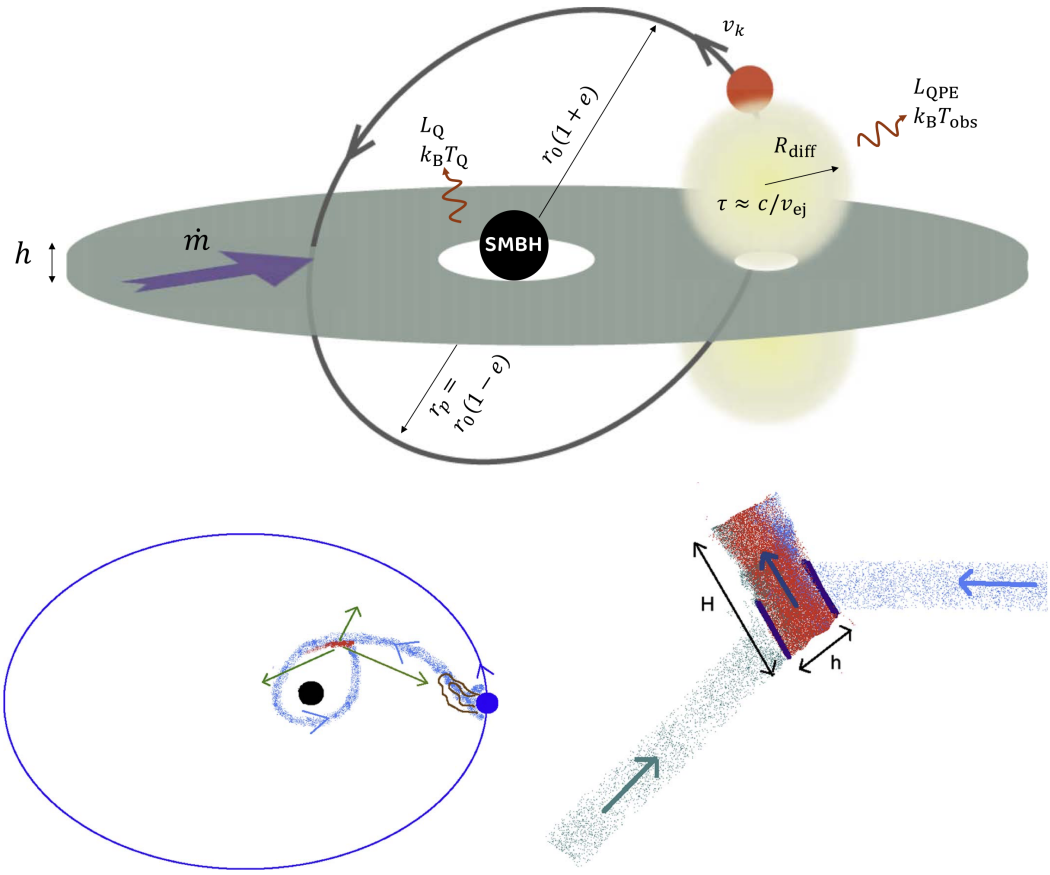

Figure 1: Examples from Linial&Metzger+23 and Krolik&Linial+22 (see both papers for more in-depth description) of an EMRI vs a disk instability model. In the EMRI model (above), the EMRI is repeatedly punching through the the disk, with the orbital shrinking from friction with the disk at the points of crossing causing the orbit's shrinking. Meanwhile, the disk instability case has a star losing its mass to the black hole, and causing shocks further in the gravitational well which cause the eruptions (by the mass transfer's non-linear nature, these shocks are quasi-periodic). GSN 069 is perfect to test the differences between the models, since the secondary object has a known optical counterpart, which means it is tidally-deformable and therefore likely to leech mass. The question this observing proposal wishes to answer is to determine which of these models dominate the EMRI signature. If the disk instability model is preferred, then tidal deformability is a must for QPEs to happen, meaning that for black holes, which the most likely EMRI secondary object, we will not have an electromagnetic signal to work with to help with parameter estimation.

Experimental Design *Describe your overall observational program. How will these observations contribute toward the accomplishment of the goals outlined in the science justification?*

We will execute a targeted and intensive monitoring campaign designed to achieve the precise timing measurements required for differentiating the two mentioned models. GSN 069 will be our target since it has a well-characterized recurrence period of $P_{\rm rec} \approx 9$ hours, a burst timing of $T_{\rm burst} \lesssim 30$ minutes. Its optical counterpart is a known Seyfert 2 galaxy (2dFGRS TGS564Z081) with magnitudes of $B = 16.2$ and $R = 15.3$, which is more than adequate for our high-cadence photometry plan with 1m telescopes. We will observe in the SDSS-r(p) band, since it is the preferred band for observing AGN activity in the literature due to optimal balancing of sensitivity and resolution. We request two primary observing epochs for our two-hour windows, each spanning around 100 hours (covering $9 - 10$ eruption cycles), separated by approximately one month: the first establishing the short-term timing precision, while the second searches for the long-term period derivative predicted by EMRI models. Getting multiple peaks in each observing run allows us to determine the average jitter difference over time, limited by the minimum jitter set by observing and modeling constraints.

The core of our strategy is to obtain high-cadence photometry around the GSN 069's predicted bursts, sampled with a cadence $T_c \approx 2$ min (100 sec exposure, 20 sec overhead) for spans of 2 hours, with the middles of these spans separated by 9 hours (the recurrence period). This will allow us to accurately sample the peak of the distribution so that the SNR analysis can be performed. To determine the target S/N for this experiment, we conducted simulations to determine the differences between the two models within our experimental design. This is depicted by figure 3, which showcases that, when a peak is offset by a certain jitter, we can then extract the SNR based on the difference between the peak of these offset peaks. In this case, the null model is the EMRI model, and the shifted model is the disk instability model, which only adds an average jitter timescale to the EMRI model. GSN 069 is in the southern sky, observable by the Cerro Tololo (Chile), the South African Astronomical Observatory site, and the Siding Spring (Australia) site of the Las Cumbres Observatory (LCO). This means, using the 0.4m telescopes at these sites, we can relate the exposure time directly to the SNR determined by the jitter. We set the lower limit of jitter to 90 seconds based on constraints from modeling the accretion flow of this object, which gives us an required detection SNR of of 145 for an exposure time of 100 seconds. This observing strategy allows us to use, for the first time, optical observations of QPE activity to constrain the physics occurring within.

References

Colless et al. 2001, *MNRAS*, 328, 1039-1063 (The 2dF Galaxy Redshift Survey: spectra and redshifts)

Shalyapin et al. 2021, *A&A*, 646, A165 (Resolving the inner accretion flow towards the central SMBH in SDSS J1339+1310)

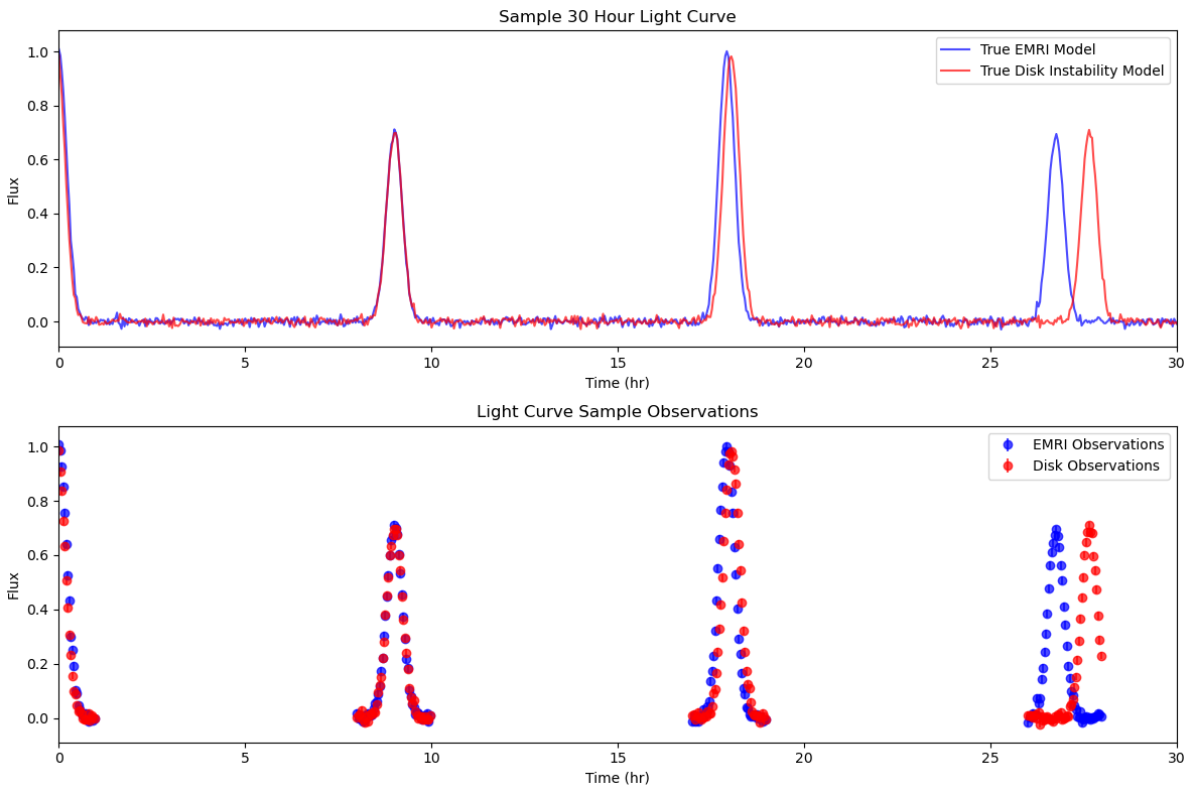

Figure 2: Example light curve and sampling plan for distinguishing the EMRI and disk instability models. The SNR required to distinguish these models is related to the jitter w.r.t. the EMRI model that the disk instability model experiences. This plot was generated with a 9 hour period, 30 minute burst time (Gaussian burst), 5 minute/period derivative, 2 hour observation windows with $\approx 30 - 50$ samples, and a 24 minute jitter in the disk instability model. Note that the sampling count and the jitter size are exaggerated to clarify the visual difference between the peaks.

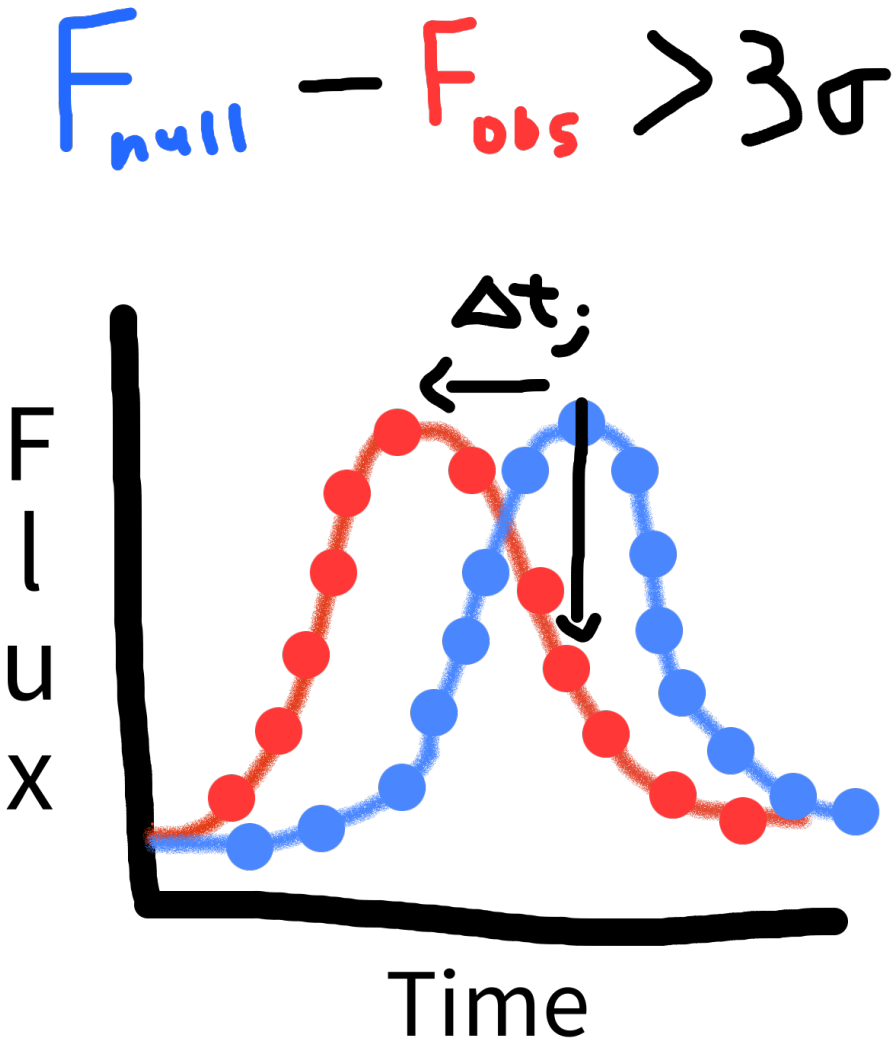

Figure 3: Visual of the observing strategy for constraining the jitter. The jitter timescale directly correlates to the difference between the null model's peak and the jitter-shifted value at the null model peak's time value. This error bar is our 3σ constraint, where that σ is then related to the null model peak's value, giving us the SNR constraint to resolve this specific jitter.